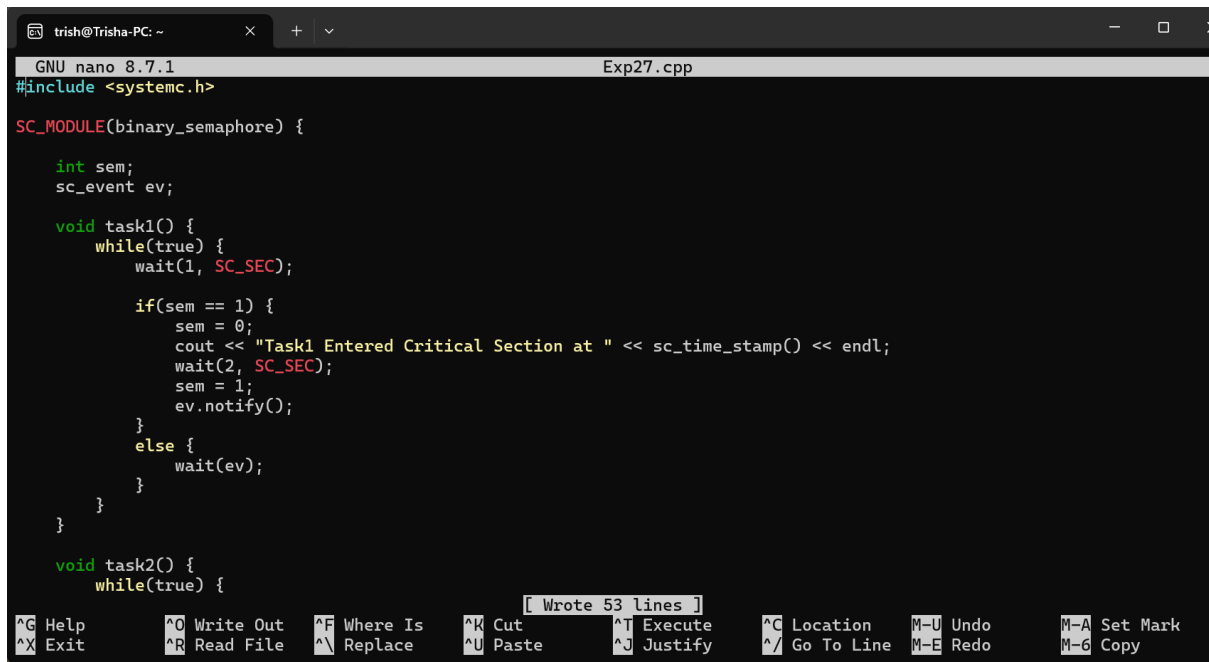


EXPERIMENT 27:



```
GNU nano 8.7.1 Exp27.cpp
#include <systemc.h>

SC_MODULE(binary_semaphore) {

    int sem;
    sc_event ev;

    void task1() {
        while(true) {
            wait(1, SC_SEC);

            if(sem == 1) {
                sem = 0;
                cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(2, SC_SEC);
                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }

    void task2() {
        while(true) {
            [ Wrote 53 lines ]
            ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
            ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-G Copy
```

```
trish@Trisha-PC:~$ nano Exp27.cpp
trish@Trisha-PC:~$ g++ -o Exp27 Exp27.cpp -lsystemc
trish@Trisha-PC:~$ ./Exp27

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
trish@Trisha-PC:~$ |
```